Experimental Probability

Name _____ **Date** _____ Calculator allowed Version 1

KEY WORDS experiment · trial · frequency · relative frequency · experimental probability · theoretical probability · fair

A Relative frequency ●○○

WORK IN YOUR BOOK

Remember: Experimental probability $= \frac{\text{number of times it happened}}{\text{number of trials}}$. This is also called the relative frequency.

1. Find the experimental probability as a fraction in its simplest form.

a) A coin is tossed 50 times and lands on heads 28 times. Find the probability of heads.

b) A dice is rolled 60 times and shows a six 12 times. Find the probability of a six.

c) A netball player scores from 18 of her 24 shots. Find the probability that she scores.

d) It rained on 9 of the 30 days in June. Find the probability of rain on a day in June.

2. Find the experimental probability as a decimal.

a) A drawing pin is dropped 200 times and lands point up 124 times. Find the probability of point up.

b) In a survey of 80 cars, 28 were white. Find the probability that a car is white.

c) A goalkeeper saves 15 of 50 shots. Find the probability of a save.

B Results in a table ●●○

WORK IN YOUR BOOK

3. A spinner with four colours is spun many times. The table shows the results.

Colour	Red	Blue	Green	Yellow
Frequency	12	18	6	4

a) How many times was the spinner spun?

b) Find the experimental probability of blue.

c) Which colour came up most often?

d) About how many times would you expect blue in 200 spins?

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C Experiment and theory ●●○

WORK IN YOUR BOOK

Remember: The theoretical probability is what should happen when every outcome is equally likely. The experimental probability is what did happen. They are usually close, but not exactly the same.

4. A fair coin is tossed 40 times. It lands on heads 23 times.

a) Find the experimental probability of heads.

b) Write down the theoretical probability of heads.

c) How many heads would you expect in 40 tosses?

d) The coin is fair. Explain why the number of heads is not exactly 20.

5. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	9	11	10	8	12	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

D Unfair dice and spinners ●●●

WORK IN YOUR BOOK

Remember: If a dice or a spinner is not fair, you cannot work out the probabilities by counting outcomes. You have to estimate them from an experiment with many trials.

6. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	6	8	7	9	5	25

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

7. A bottle top is flipped 250 times. It lands the right way up 90 times.

a) Estimate the probability that it lands the right way up.

b) About how many times would it land the right way up in 1,000 flips?

★ Extension

FOR EARLY FINISHERS

8. Ana tosses a coin 10 times and gets 7 heads. Ben tosses the same coin 500 times and gets 262 heads. Find each experimental probability of heads as a decimal. Whose result tells you more about whether the coin is fair? Explain.

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Experimental Probability

Name _____ Date _____ Calculator allowed Version 2

KEY WORDS experiment · trial · frequency · relative frequency · experimental probability · theoretical probability · fair

A Relative frequency ●○○

WORK IN YOUR BOOK

Remember: Experimental probability $= \frac{\text{number of times it happened}}{\text{number of trials}}$. This is also called the relative frequency.

1. Find the experimental probability as a fraction in its simplest form.

a) A coin is tossed 40 times and lands on heads 22 times. Find the probability of heads.

b) A dice is rolled 90 times and shows a six 18 times. Find the probability of a six.

c) A netball player scores from 21 of her 28 shots. Find the probability that she scores.

d) It rained on 12 of the 30 days in June. Find the probability of rain on a day in June.

2. Find the experimental probability as a decimal.

a) A drawing pin is dropped 200 times and lands point up 116 times. Find the probability of point up.

b) In a survey of 80 cars, 36 were white. Find the probability that a car is white.

c) A goalkeeper saves 20 of 50 shots. Find the probability of a save.

B Results in a table ●●○

WORK IN YOUR BOOK

3. A spinner with four colours is spun many times. The table shows the results.

Colour	Red	Blue	Green	Yellow
Frequency	20	10	6	4

a) How many times was the spinner spun?

b) Find the experimental probability of red.

c) Which colour came up most often?

d) About how many times would you expect red in 200 spins?

C Experiment and theory ●●○

WORK IN YOUR BOOK

Remember: The theoretical probability is what should happen when every outcome is equally likely. The experimental probability is what did happen. They are usually close, but not exactly the same.

4. A fair coin is tossed 60 times. It lands on heads 27 times.

a) Find the experimental probability of heads.

b) Write down the theoretical probability of heads.

c) How many heads would you expect in 60 tosses?

d) The coin is fair. Explain why the number of heads is not exactly 30.

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5. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	11	9	10	12	8	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

D Unfair dice and spinners ●●●

WORK IN YOUR BOOK

> **Remember:** If a dice or a spinner is not fair, you cannot work out the probabilities by counting outcomes. You have to estimate them from an experiment with many trials.

6. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	7	5	8	6	9	25

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

7. A bottle top is flipped 250 times. It lands the right way up 110 times.

a) Estimate the probability that it lands the right way up.

b) About how many times would it land the right way up in 1,000 flips?

Extension

FOR EARLY FINISHERS

8. Ana tosses a coin 10 times and gets 3 heads. Ben tosses the same coin 500 times and gets 241 heads. Find each experimental probability of heads as a decimal. Whose result tells you more about whether the coin is fair? Explain.

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